BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Technology-Assisted Depression Screening Tools for Cancer Patients: A Systematic Review Protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041878
Article Type:	Protocol
Date Submitted by the Author:	07-Aug-2020
Complete List of Authors:	Lazaro-Escudero, Maria; Ponce Health Sciences University, Burgos-Cardona, Camila; Ponce Health Sciences University Acevedo-Fernández, Karina; Ponce Health Sciences University Castro , EM ; Ponce Health Sciences University,
Keywords:	ONCOLOGY, MENTAL HEALTH, Depression & mood disorders < PSYCHIATRY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Title: Technology-Assisted Depression Screening Tools for Cancer Patients: A Systematic Review Protocol

María Isabel Lázaro-Escudero, BS[a]
mlazaro17@stu.psm.edu
Camila Alanna Burgos-Cardona, MS[a]
cburgos17@stu.psm.edu
Karina Acevedo-Fernández, BA[a]
kacevedo15@stu.psm.edy
Eida M. Castro-Figueroa, PsyD, MS[a,b]
ecastro@psm.edu

Key words: Depression, Cancer, technology, screening

Correspondence to: Eida M. Castro-Figueroa;
ecastro@psm.edu;
388 Zona Industrial Reparada 2,
Ponce, PR. 00716.

For peer review only

ABSTRACT

Introduction: Among cancer patients, depression is still under-detected. The use of technology-assisted screening tools is rising; however, little is known about the uptake of these devices as depression screening tools among cancer patients.

Methods and analysis: A systematic review will be conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P). The review is registered with PROSPERO and any adjustments to the protocol will be traced. The aims of this systematic review are to (1) identify the most common and feasible depression screening information technology (IT) delivery models among cancer patients (2) identify the most common depression screening instrument used in IT devices and (3) describe the published technology-assisted depression screening tools for cancer patients. PubMed, EBSCOhost and Google Scholar databases will be used. PICO guidelines will inform the inclusion criteria. Two researchers will independently review titles and abstracts, followed by full article review and data extraction. In the case of a disagreement, a third reviewer will make the final decision. Title/abstract screening will be conducted using a screening tool prepared by the researchers. Articles will be included for review if: 1) the study includes cancer patients, cancer survivors, and/or patients on remission, 2) depression is screened using technology and 3) technology-assisted depression screening effectiveness, efficacy, feasibility and/or acceptance is addressed. The quality of the articles will be assessed using the Methodological Index For Non-Randomized Studies (MINORS, maximum score 24) through independent coding of reviewers.

For peer review only

Ethics and Dissemination: This research is exempt from ethics approval given that this is a protocol for a systematic review, which uses published data. Findings from this review will be disseminated through peer-reviewed publications and scientific conferences.

Systematic review registration: PROSPERO registration number CRD42019121048.

Strengths and limitations of this study:

- This systematic review addresses the use of technology for depression screening in cancer patients, as well as the process evaluation (feasibility, acceptability, efficacy and/or effectiveness) of IT delivery models in oncology practice.
- Article screening, data extraction, quality and strength of evidence assessment will be independently conducted by two reviewers to reduce personal bias.
- Studies in languages other than English will not be included or searched, which may contribute to language bias.

INTRODUCTION

Rationale

More than 300 million people worldwide suffer from depression.[1] Depressive spectrum disorders seem to be a common mental health complication of cancer.[2] A study meta-analysis reported the prevalence of depression symptoms in cancer patients to range from 8%-24%, varying due to type of screening instrument, type of cancer, and treatment phase.[3] A large epidemiological multi-center study found that cancer patients have a five-fold increased chance of experiencing depression in comparison to the general population.[4] Studies investigating the

For peer review only

association between cancer treatments and depression have found a higher prevalence of depressive disorders in patients in active treatment compared to those who are not receiving treatment.[2] If these numbers are analyzed in the context that globally 18.1 million new cases of cancer were identified in 2018, [1, 5] screening depression symptoms among cancer patients should be a health priority.

The screening of depression is now becoming a significant aspect of cancer care.[6] For example, the National Comprehensive Cancer Network and the Institute of Medicine have created guidelines and recommendations for routine screening of distress (including depression symptoms) as a quality standard in oncology practice.[7] It is recommended that those responsible for the care of patients with cancer screen for depression and provide further assessment for patients that screen positive for depression; and the process should identify patients that could benefit from additional support.[8] It has been demonstrated that distress screening improves the management of psychosocial distress in cancer patients.[9] Depression symptoms remain under-recognized among cancer patients and as such they are associated with major effects on patient suffering, mortality and healthcare expenditure.[10] Despite the existence of evidence-based recommendations to screen and address depression symptoms, depression continues to go under-detected and under-treated [11,12]. One of the main causes of this lack of implementation of depression screening is thought to be limited resources and the proper implementation of this service in most government hospitals.[13]

The use of technology-assisted depression screening tools is beginning to be acknowledged by the US National Institutes of Health (NIH).[14] Compared to the traditional paper-pencil method, electronic devices for symptom assessment have proven to offer diverse benefits such as immediate presentation of results and compilation of aggregate scores, as well as the possibility

For peer review only

of tracking symptoms.[15] The use of screening technology potentiates mental healthcare efficiency by enabling easy access to in-depth follow-up assessment, referral and treatment.[16] There has been an increased implementation of technology-assisted tools in the cancer care system.[17, 18, 19, 20] The health care system has not taken full advantage of the benefits that major advances in computer technology would provide in aiding routine oncology practice.[21]

Objectives

Little is known about the extent to which IT devices are being incorporated into routine depression screening initiatives among cancer patients. Moreover, we are not aware of any systematic reviews addressing the use of technology for depression screening in cancer patients. The purposes of this systematic review are to:

1. identify the most common delivery models of depression screening information technology (IT) among cancer patients;
2. identify the most common instrument used to screen depression in IT devices.
3. describe the published technology-assisted depression screening tools and explore its feasibility for cancer patients.

METHODS AND ANALYSIS

Eligibility Criteria

Study characteristics

Inclusion criteria

PICO[22] guidelines were used to develop inclusion criteria. Articles will be included if they focus on feasibility, acceptability, efficacy or effectiveness of the use of technology assisted tools to facilitate screening of depression symptoms among cancer patients seeking medical care (PICO-context). Articles may include those with randomized-control design as long as the comparison

For peer review only

group is exposed to other technology assisted screening tools or the paper and pencil screening format. Also, the protocol includes scientific articles addressing IT-assisted depression screening's feasibility, acceptability and implementation of among cancer patients. In summary, articles with the following criteria will be included for review:

- Experimental or quasi-experimental research design;
- Randomized and non-randomized control trials;
- Studies involving cancer patients, cancer-free subjects and/or patients on remission;
- Studies using a technology-assisted depression screening tool;
- Studies evaluating the effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool;
- Articles written in English

Exclusion criteria

Articles with the following criteria will be excluded for review:

- Studies in which screening technology is used to measure depression prevalence;
- Studies incorporating the use of information technology devices to facilitate baseline and follow up assessments of depression symptoms;
- Studies using technology devices to deliver behavioral interventions to treat depression;
- Meta-analyses.

Patient and Public Involvement

No patients or public will be involved in this study considering it is a systematic review.

For peer review only

Information sources

Electronic databases

The following electronic databases will be searched: PubMed, EBSCOhost and Google Scholar.

Additional sources

Articles will also be obtained from the following sources:

- Reference list of included studies;
- Reference list of an excluded study if it shows direct relevance to fit inclusion criteria;
- Manual searching of articles.

Search strategy

Search strategy for PubMed is in online supplementary appendix 1. This search strategy will be adapted to the other databases, reporting further modifications in the review manuscript.

Study selection

Two main reviewers will independently screen the title and abstract of the articles resulting from keyword search in the three databases. Screening of articles will be made using a standardized screening tool, containing the year the study was published, the first author's last name, checkmarks for inclusion criteria, and acceptance/rejection of the study. The full text of manuscripts will be obtained if the title and abstract content does not clearly state the inclusion/exclusion criteria. Discrepancies between the two main reviewers will be resolved through discussion and consensus. If there is still discrepancy after discussion, a third reviewer will be added. The third reviewer's selection criteria will also be discussed with the two main reviewers. Duplicates will be removed using the Excel sheet search icon. The criteria for selection or exclusion of all articles will be documented.

For peer review only

Data extraction

The two review authors will independently extract data from the included studies using an Excel chart that will take into account the inclusion and exclusion criteria (Table 1). A practice data extraction section will be performed by reviewers before conducting the data extraction. Any disagreement regarding data extraction will be resolved through discussion and consensus. A third reviewer will resolve any discrepancies.

Table 1

First Author	Year	Title	Cancer patients, cancer-free subjects and/or patients on remission.	Depression or distress screening using technology.	Experimental or quasi-experimental research design.	IT screening delivery model effectiveness, efficacy, feasibility and/or acceptance is addressed.	Articles written in English.

Data items

The following information will be extracted:

- Articles' full bibliographical data;
- Study site, screening instrument, type of technology delivery model, primary and secondary outcomes;
- Sample characteristics and demographics, including sample size, cancer type, nationality and mean age.
- Number of participants, including number of participants per arm in randomized control trials;
- Study methodology, recruitment site and study completion rates;
- Source(s) of research funding and potential conflicts of interest.

For peer review only

Risk of bias (quality) assessment

The quality of methods in the study will be assessed with the Methodological Index For Non-Randomized Studies (MINORS)[23]. Two (2) independent coders will review all articles included after inclusion/exclusion criteria are applied. Any disagreements will be resolved by discussion that will include a third (3) reviewer. The MINORS tool integrates two levels of assessment, one for quasi-experimental and one for RCTs. The following items will be considered for bias assessment:

1. A clearly stated aim
2. Inclusion of consecutive patients
3. Prospective collection of data
4. Endpoints appropriate to the aim of the study
5. Unbiased assessment of the study endpoint
6. Follow-up period appropriate to the aim of the study
7. Loss to follow up less than 5%
8. Prospective calculation of the study size

Additional criteria in the case of comparative study:

9. An adequate control group
10. Contemporary groups
11. Baseline equivalence of groups
12. Adequate statistical analyses

Data analysis

Descriptive analysis will be performed to assess frequency and percentage of IT technology tools used for depression screening among cancer patients, participants' types of cancer and

For peer review only

demographic characteristics (sex, age, ethnicity, income). Also, we will calculate the mean age of overall participants. Moreover, we will perform a narrative synthesis of main outcomes based on the extracted data items (Sample Table 2).

Sample Table 2

Article	Study site	Year published	Number of participants	Sample characteristics and demographics	Screening instrument	Type of technology delivery model	Primary and secondary outcomes	Study methodology

Grading the strength of evidence

We will use the GRADE (Grades of Recommendation, Assessment, Development and Evaluation) approach to assess the strength of evidence of the studies reviewed, as recommended by the Cochrane Handbook for Systematic Reviews of Interventions.[24] This approach evaluates risk of bias, imprecision, inconsistency, indirectness and publication bias. The quality of evidence is rated at all levels: very low, low, moderate and high.

ETHICS AND DISSEMINATION

Results of this study will be disseminated in scientific conferences and as a peer-reviewed scientific publication. Since the primary data will be extracted from public domain databases, no informed consent of formal ethics committee review is required.

DISCUSSION

The National Comprehensive Cancer Network and the Institute of Medicine[25] have created guidelines and recommendations for routine screening of distress (including depression symptoms) as a quality standard in oncology practice. Despite the existence of such evidence-

For peer review only

based recommendations[26], depression continues to go under-detected and under- treated. The conclusions drawn by this systematic approach describe progress towards the development of new technology-assisted tools for depression screening in standard oncology practice.

Conflicts of interest: None declared.

Author Affiliation

[a]School of Behavioral and Brain Sciences, Ponce Research Institute, Ponce Health Sciences University.

[b]Department of Psychiatry, Ponce Research Institute, Ponce Health Sciences University.

Contributions

EC is the guarantor of this review. ML and EC conceptualized the systematic review protocol. ML and CB, drafted the manuscript, and ML, EC, CB and KA reviewed the protocol critically.

Funding

This work was supported by the National Cancer Institute (2U54CA163071 and 2U54CA163068) and the National Institute of Minority Health and Health Disparities (5G12MD007579-33, 5R25MD007607, R21MD013674 and 2U54MD007579-34).

References

1. World Health Organization (WHO): Depression (2018). Retrieved from https://www.who.int/en/news-room/fact-sheets/detail/depression
2. Caruso R, Nanni MG, Riba M, et al. (2017) Depressive spectrum disorders in cancer: prevalence, risk factors and screening. A critical review Acta Oncologica. **56**:146–55.

For peer review only

3. Krebber AM, Buffart LM, Kleijn G, et al. (2014) Prevalence of depression in cancer patients: a meta-analysis of diagnostic interviews and self-report instruments. *Psychooncology*. **23**:121-130.
4. Hartung, T. J., Brähler, E., Faller, H., Härter, M., Hinz, A., Johansen, C., Mehnert, A. (2017). The risk of being depressed is significantly higher in cancer patients than in the general population: Prevalence and severity of depressive symptoms across major cancer types. *European Journal of Cancer, 72, 46–53.*doi:10.1016/j.ejca.2016.11.017
5. World Health Organization (WHO): Cancer. (2018) Retrieved from https://www.who.int/cancer/en/
6. Penedo, F. J., & Cella, D. (2016). Responding to the quality imperative to embed mental health care into ambulatory oncology. *Cancer, 123(3), 382–386.*doi:10.1002/cncr.30402
7. Lowery AE, Holland JC. (2011) Screening cancer patients for distress: Guidelines for routine implementation. Community Oncol. **8**: 502–5.
8. Smith, H. R. (2015). Depression in cancer patients: Pathogenesis, implications and treatment. *Oncology letters*, *9*(4), 1509-1514.
9. Howell D., Alici Y. (2010) Management of distress in cancer patients. *J Support Oncol*. 8: 4-12.
10. Mausbach, B. T., & Irwin, S. A. (2017). Depression and healthcare service utilization in patients with cancer. *Psycho-oncology*, *26*(8), 1133-1139.
11. Dauchy, S., Dolbeault, S., & Reich, M. (2013). Depression in cancer patients. *EJC Supplements*, *11*(2), 205.

For peer review only

12. Walker, J., Hansen, C. H., Martin, P., et al. (2014). Prevalence, associations, and adequacy of treatment of major depression in patients with cancer: a cross-sectional analysis of routinely collected clinical data. *The Lancet Psychiatry*, *1*(5), 343-350.

13. Vaibhav Kumar Ingle, Ijya Pandey, Akash Ranjan Singh, et al. (2017) Screening of Patients with Chronic Medical Disorders in the Outpatient Department for Depression Using Handheld Computers as Interface and Patient Health Questionnaire-9 as a Tool. Int J Appl Basic Med Res. 7(2): 129–133. doi: 10.4103/2229-516X.205809

14. "PA-18-493: Using Information Technology to Support Systematic Screening and Treatment of Depression in Cancer (R01 Clinical Trial Optional)." *National Institutes of Health*, U.S. Department of Health and Human Services, grants.nih.gov/grants/guide/pa-files/pa-18-493.html.

15. M.J. Hjermstad, H.C. Lie, A. Caraceni, *et al. (2012)* Computer-based symptom assessment is feasible in patients with advanced cancer: results from an international multicenter study, the EPCRC-CSA. J Pain Symptom Manage. 44; 639-654

16. Kingston D, McDonald S, Biringer A, et al. (2014) Comparing the feasibility, acceptability, clinical-, and cost-effectiveness of mental health e-screening to paper-based screening on the detection of depression, anxiety, and psychosocial risk in pregnant women: a study protocol of a randomized, parallel-group, superiority trial. Trials. 15:3. doi: 10.1186/1745-6215-15-3.

17. Aktas A, Hullihen B, Shrotriya S, Thomas S, Walsh D, et al. (2013) Connected Health: Cancer Symptom and Quality-of-Life Assessment Using a Tablet Computer: A Pilot Study. Am J Hosp Palliat Care. doi:10.1177/1049909113510963.

For peer review only

18. Gell NM, Grover KW, Humble M , *et al*. (2017) Efficacy, feasibility, and acceptability of a novel technology-based intervention to support physical activity in cancer survivors. *Support Care Cancer*2017;25:1291–300.doi:10.1007/s00520-016-3523-5

19. Quintiliani LM, Mann DM, Puputti M, Quinn E, Bowen DJ. Pilot and feasibility test of a mobile health-supported behavioral counseling intervention for weight management among breast cancer survivors. JMIR Cancer. 2016;2(1):e4.

20. Stinson, J. N., Jibb, L. A., Nguyen, C., Nathan, P. C., Maloney, A. M., Dupuis, L. L., et al. (2015). Construct validity and reliability of a real-time multidimensional smartphone app to assess pain in children and adolescents with cancer. *Pain, 156*(12), 2607–2615. doi: 10.1097/j.pain.0000000000000385.

21. Christ, G. H., Messner, C., & Behar, L. C. (2015). *Handbook of oncology social work: Psychosocial care of the patient with cancer*. New York: Oxford University Press.

22. Using PICO or PICo. (2019) Retrieved from https://libguides.murdoch.edu.au/systematic/PICO.

23. Dent, O. (2003). Methodological index for non-randomized studies. *ANZ Journal of Surgery,73*(9), 675-676. doi:10.1046/j.1445-2197.2003.02762.x

24. Higgins, J. P. T. (2020). Cochrane handbook for systematic reviews of interventions. Hoboken, NJ: Wiley-Blackwell.

25. Holland JC, Andersen B, Breitbart WS, et al. Distress Management. *J Natl Compr Cancer Netw*. 2013;11:190-2009. doi:10.6004/jnccn.2013.0027.

26. Mitchell AJ, Vahabzadeh A, Magruder K. Screening for distress and depression in cancer settings: 10 lessons from 40 years of primary-care research. *Psychooncology*. 2011;20(6):572-584. doi:10.1002/pon.1943.

For peer review only

Pubmed Search Strategy

((((Distress[Title/Abstract] OR depression[Title/Abstract])) AND screening) AND technology) AND cancer

((("Distress"[Title/Abstract] OR "depression"[Title/Abstract]) AND ((((((((((("diagnosis"[MeSH Subheading] OR "diagnosis"[All Fields]) OR "screening"[All Fields]) OR "mass screening"[MeSH Terms]) OR ("mass"[All Fields] AND "screening"[All Fields])) OR "mass screening"[All Fields]) OR "early detection of cancer"[MeSH Terms]) OR (("early"[All Fields] AND "detection"[All Fields]) AND "cancer"[All Fields])) OR "early detection of cancer"[All Fields]) OR "screen"[All Fields]) OR "screenings"[All Fields]) OR "screened"[All Fields]) OR "screens"[All Fields])) AND (((("technology"[MeSH Terms] OR "technology"[All Fields]) OR "technologies"[All Fields]) OR "technology s"[All Fields])) AND (((((((("cancer s"[All Fields] OR "cancerated"[All Fields]) OR "canceration"[All Fields]) OR "cancerization"[All Fields]) OR "cancerized"[All Fields]) OR "cancerous"[All Fields]) OR "neoplasms"[MeSH Terms]) OR "neoplasms"[All Fields]) OR "cancer"[All Fields]) OR "cancers"[All Fields])

Translations

screening: "diagnosis"[Subheading] OR "diagnosis"[All Fields] OR "screening"[All Fields] OR "mass screening"[MeSH Terms] OR ("mass"[All Fields] AND "screening"[All Fields]) OR "mass screening"[All Fields] OR "early detection of cancer"[MeSH Terms] OR ("early"[All Fields] AND "detection"[All Fields] AND "cancer"[All Fields]) OR "early detection of cancer"[All Fields] OR "screen"[All Fields] OR "screenings"[All Fields] OR "screened"[All Fields] OR "screens"[All Fields]

technology: "technology"[MeSH Terms] OR "technology"[All Fields] OR "technologies"[All Fields] OR "technology's"[All Fields]

cancer: "cancer's"[All Fields] OR "cancerated"[All Fields] OR "canceration"[All Fields] OR "cancerization"[All Fields] OR "cancerized"[All Fields] OR "cancerous"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields] OR "cancers"[All Fields]

For peer review only

PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Page number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	NA
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	3
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☐	NA
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐	☐	NA
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	13
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☐	NA
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	3-5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	5
METHODS					

For peer review only

BioMed Central The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Page number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	5-6
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	7 (Supplemental File 1)
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	7
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	7
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	7
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	9-10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	8-9
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	9-10
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	9-10
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☐	NA
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☐	NA
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☐	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	10

For peer review only

BioMed Central
The Open Access Publisher

BMJ Open

Technology-Assisted Depression Screening Tools for Cancer Patients: A Systematic Review Protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041878.R1
Article Type:	Protocol
Date Submitted by the Author:	24-Dec-2020
Complete List of Authors:	Lazaro-Escudero, Maria; Ponce Health Sciences University, Burgos-Cardona, Camila; Ponce Health Sciences University Acevedo-Fernández, Karina; Ponce Health Sciences University Castro , EM ; Ponce Health Sciences University,
Primary Subject Heading:	Oncology
Secondary Subject Heading:	Mental health
Keywords:	ONCOLOGY, MENTAL HEALTH, Depression & mood disorders < PSYCHIATRY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Title: Technology-Assisted Depression Screening Tools for Cancer Patients: A Systematic Review Protocol

María Isabel Lázaro-Escudero, BS[a]
mlazaro17@stu.psm.edu
Camila Alanna Burgos-Cardona, MS[a]
cburgos17@stu.psm.edu
Karina Acevedo-Fernández, BA[a]
kacevedo15@stu.psm.edy
EM Castro, PsyD, MS[a,b]
ecastro@psm.edu

Key words: Depression, Cancer, technology, screening

Correspondence to: Eida M. Castro-Figueroa;
ecastro@psm.edu;
388 Zona Industrial Reparada 2,
Ponce, PR. 00716.

For peer review only

ABSTRACT

Introduction: Among cancer patients, depression is still under-detected. The use of technology-assisted screening tools is rising; however, little is known about the uptake of these devices as depression screening tools among cancer patients.

Methods and analysis: A systematic review will be conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P). The review is registered with PROSPERO and any adjustments to the protocol will be traced. The aims of this systematic review are to (1) identify the most common and feasible depression screening information technology (IT) delivery models among cancer patients (2) identify the most common depression screening instrument used in IT devices and (3) describe the published technology-assisted depression screening tools for cancer patients. PubMed, EBSCOhost and Google Scholar databases will be used. PICO guidelines will inform the inclusion criteria. Two researchers will independently review titles and abstracts, followed by full article review and data extraction. In the case of a disagreement, a third reviewer will make the final decision. Title/abstract screening will be conducted using a screening tool prepared by the researchers. Articles will be included for review if: 1) the study includes cancer patients, cancer survivors, and/or patients on remission, 2) depression is screened using technology and 3) technology-assisted depression screening effectiveness, efficacy, feasibility and/or acceptance is addressed. The quality of the articles will be assessed using the Methodological Index For Non-Randomized Studies (MINORS, maximum score 24) through independent coding of reviewers.

For peer review only

Ethics and Dissemination: This research is exempt from ethics approval given that this is a protocol for a systematic review, which uses published data. Findings from this review will be disseminated through peer-reviewed publications and scientific conferences.

Systematic review registration: PROSPERO registration number CRD42019121048.

Strengths and limitations of this study:

- This systematic review addresses the use of technology for depression screening in cancer patients, as well as the process evaluation (feasibility, acceptability, efficacy and/or effectiveness) of IT delivery models in oncology practice.
- Article screening, data extraction, quality and strength of evidence assessment will be independently conducted by two reviewers to reduce personal bias.
- Studies in languages other than English will not be included or searched, which may contribute to language bias.

INTRODUCTION

Rationale

More than 300 million people worldwide suffer from depression.[1] Depressive spectrum disorders seem to be a common mental health complication of cancer.[2] A study meta-analysis reported the prevalence of depression symptoms in cancer patients to range from 8%-24%, varying due to type of screening instrument, type of cancer, and treatment phase.[3] A large epidemiological multi-center study found that cancer patients have a five-fold increased chance of experiencing depression in comparison to the general population.[4] Studies investigating the

For peer review only

association between cancer treatments and depression have found a higher prevalence of depressive disorders in patients in active treatment compared to those who are not receiving treatment.[2] If these numbers are analyzed in the context that globally 18.1 million new cases of cancer were identified in 2018, [1, 5] screening depression symptoms among cancer patients should be a health priority.

The screening of depression is now becoming a significant aspect of cancer care.[6] For example, the National Comprehensive Cancer Network and the Institute of Medicine have created guidelines and recommendations for routine screening of distress (including depression symptoms) as a quality standard in oncology practice.[7] It is recommended that those responsible for the care of patients with cancer screen for depression and provide further assessment for patients that screen positive for depression; and the process should identify patients that could benefit from additional support.[8] It has been demonstrated that distress screening improves the management of psychosocial distress in cancer patients.[9] Depression symptoms remain under-recognized among cancer patients and as such they are associated with major effects on patient suffering, mortality and healthcare expenditure.[10] Despite the existence of evidence-based recommendations to screen and address depression symptoms, depression continues to go under-detected and under-treated [11,12]. One of the main causes of this lack of implementation of depression screening is thought to be limited resources and the proper implementation of this service in most government hospitals.[13]

The use of technology-assisted depression screening tools is beginning to be acknowledged by the US National Institutes of Health (NIH).[14] Compared to the traditional paper-pencil method, electronic devices for symptom assessment have proven to offer diverse benefits such as immediate presentation of results and compilation of aggregate scores, as well as the possibility

For peer review only

of tracking symptoms.[15] The use of screening technology potentiates mental healthcare efficiency by enabling easy access to in-depth follow-up assessment, referral and treatment.[16] There has been an increased implementation of technology-assisted tools in the cancer care system.[17, 18, 19, 20] The health care system has not taken full advantage of the benefits that major advances in computer technology would provide in aiding routine oncology practice.[21]

Objectives

Little is known about the extent to which information technology (IT) devices (computers, tablets, mobile phones, etc.) are being incorporated into routine depression screening initiatives among cancer patients. Moreover, we are not aware of any systematic reviews addressing the use of technology for depression screening in cancer patients. The purposes of this systematic review are to:

1. identify the most common delivery models of depression screening IT (internet browser, type of platform, application software etc.) among cancer patients;
2. identify the most common instrument used to screen depression in IT devices.
3. describe the published technology-assisted depression screening tools and explore its feasibility for cancer patients.

METHODS AND ANALYSIS

Eligibility Criteria

Study characteristics

Inclusion criteria

PICO[22] guidelines were used to develop inclusion criteria. Articles will be included if they focus on feasibility, acceptability, efficacy or effectiveness of the use of technology assisted tools to facilitate screening of depression symptoms among cancer patients seeking medical care (PICO-

For peer review only

context). Articles may include those with randomized-control design as long as the comparison group is exposed to other technology assisted screening tools or the paper and pencil screening format. Also, the protocol includes scientific articles addressing IT-assisted depression screening's feasibility, acceptability and implementation of among cancer patients. In summary, articles with the following criteria will be included for review:

- Experimental or quasi-experimental research design;
- Randomized and non-randomized control trials;
- Studies involving cancer patients, cancer-free subjects and/or patients on remission;
- Studies using a technology-assisted depression screening tool;
- Studies evaluating the effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool;
- Articles written in English

Exclusion criteria

Articles with the following criteria will be excluded for review:

- Studies in which screening technology is used to measure depression prevalence;
- Studies using technology devices to deliver behavioral interventions to treat depression;
- Studies not defining criteria for the measurement of effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool;
- Studies using only qualitative assessments to evaluate effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool;
- Meta-analyses.

For peer review only

Patient and Public Involvement

No patients or public will be involved in this study considering it is a systematic review.

Information sources

Electronic databases

The following electronic databases will be searched: PubMed, EBSCOhost and Google Scholar.

Additional sources

Articles will also be obtained from the following sources:

- Reference list of included studies;
- Reference list of an excluded study if it shows direct relevance to fit inclusion criteria;
- Manual searching of articles.

Search strategy

Search strategy for PubMed is in online supplementary appendix 1. This search strategy will be adapted to the other databases, reporting further modifications in the review manuscript. Articles published up to December 2020 will be included in this systematic review.

Study selection

Two main reviewers will independently screen the title and abstract of the articles resulting from keyword search in the three databases. Screening of articles will be made using a standardized screening tool, containing the year the study was published, the first author's last name, checkmarks for inclusion criteria, and acceptance/rejection of the study. The full text of manuscripts will be obtained if the title and abstract content does not clearly state the inclusion/exclusion criteria. Discrepancies between the two main reviewers will be resolved through discussion and consensus. If there is still discrepancy after discussion, a third reviewer

For peer review only

will be added. The third reviewer's selection criteria will also be discussed with the two main reviewers. Duplicates will be removed using the Excel sheet search icon. The criteria for selection or exclusion of all articles will be documented.

Data extraction

The two review authors will independently extract data from the included studies using an Excel chart that will take into account the inclusion and exclusion criteria (Table 1). A practice data extraction section will be performed by reviewers before conducting the data extraction. Any disagreement regarding data extraction will be resolved through discussion and consensus. A third reviewer will resolve any discrepancies.

Table 1: Data Extraction Items Based on Inclusion Criteria

First Author	Year	Title	Cancer patients, cancer-free subjects and/or patients on remission.	Depression or distress screening using technology.	Experimental or quasi-experimental research design.	IT screening delivery model effectiveness, efficacy, feasibility and/or acceptance is addressed.	Articles written in English.

Data items

The following information will be extracted:

- Articles' full bibliographical data;
- Model of service delivery type (oncology practice, primary care setting, inpatient setting, etc.), screening instrument, type of technology delivery model, primary and secondary outcomes, screening process (with or without confirmed depression diagnosis);
- Sample characteristics and demographics, including sample size, cancer type, nationality and mean age.

For peer review only

- Number of participants, including number of participants per arm in randomized control trials;
- Study methodology, recruitment site and study completion rates;
- Source(s) of research funding and potential conflicts of interest;
- The following parameters will be pooled from each study selected: percent of 'relapse patients', study design (including exact time points when longitudinal), scale scoring criteria (including cut-off values).

Risk of bias (quality) assessment

The quality of methods in the study will be assessed with the Methodological Index For Non-Randomized Studies (MINORS)[23]. Two (2) independent coders will review all articles included after inclusion/exclusion criteria are applied. Any disagreements will be resolved by discussion that will include a third (3) reviewer. The MINORS tool integrates two levels of assessment, one for quasi-experimental and one for RCTs. The following items will be considered for bias assessment:

1. A clearly stated aim
2. Inclusion of consecutive patients
3. Prospective collection of data
4. Endpoints appropriate to the aim of the study
5. Unbiased assessment of the study endpoint
6. Follow-up period appropriate to the aim of the study
7. Loss to follow up less than 5%
8. Prospective calculation of the study size

For peer review only

Additional criteria in the case of comparative study:

9. An adequate control group
10. Contemporary groups
11. Baseline equivalence of groups
12. Adequate statistical analyses

Data analysis

Descriptive analysis will be performed to assess frequency and percentage of IT technology tools used for depression screening among cancer patients, participants' types of cancer and demographic characteristics (sex, age, ethnicity, income). Also, we will calculate the mean age of overall participants. Moreover, we will perform a narrative synthesis of main outcomes based on the extracted data items (Sample Table 2).

Sample Table 2: Characteristics of Included Articles and Outcomes

Article	Model of service delivery	Year published	Number of participants	Sample characteristics and demographics	Screening instrument	Type of technology delivery model	Primary and secondary outcomes	Study methodology

Grading the strength of evidence

We will use the GRADE (Grades of Recommendation, Assessment, Development and Evaluation) approach to assess the strength of evidence of the studies reviewed, as recommended by the Cochrane Handbook for Systematic Reviews of Interventions.[24] This approach evaluates risk of bias, imprecision, inconsistency, indirectness and publication bias. The quality of evidence is rated at all levels: very low, low, moderate and high.

For peer review only

ETHICS AND DISSEMINATION

Results of this study will be disseminated in scientific conferences and as a peer-reviewed scientific publication. Since the primary data will be extracted from public domain databases, no informed consent of formal ethics committee review is required.

DISCUSSION

The National Comprehensive Cancer Network and the Institute of Medicine[25] have created guidelines and recommendations for routine screening of distress (including depression symptoms) as a quality standard in oncology practice. Despite the existence of such evidence-based recommendations[26], depression continues to go under-detected and under- treated. The conclusions drawn by this systematic approach describe progress towards the development of new technology-assisted tools for depression screening in standard oncology practice.

Conflicts of interest: None declared.

Author Affiliation

[a]School of Behavioral and Brain Sciences, Ponce Research Institute, Ponce Health Sciences University.

[b]Department of Psychiatry, Ponce Research Institute, Ponce Health Sciences University.

Contributions

EC is the guarantor of this review. ML and EC conceptualized the systematic review protocol. ML and CB, drafted the manuscript, and ML, EC, CB and KA reviewed the protocol critically.

Funding

This work was supported by the National Cancer Institute (2U54CA163071 and 2U54CA163068) and the National Institute of Minority Health and Health Disparities (5G12MD007579-33, 5R25MD007607, R21MD013674 and 2U54MD007579-34).

For peer review only

References

1. World Health Organization (WHO): Depression (2018). Retrieved from https://www.who.int/en/news-room/fact-sheets/detail/depression
2. Caruso R, Nanni MG, Riba M, et al. (2017) Depressive spectrum disorders in cancer: prevalence, risk factors and screening. A critical review Acta Oncologica. **56**:146–55.
3. Krebber AM, Buffart LM, Kleijn G, et al. (2014) Prevalence of depression in cancer patients: a meta-analysis of diagnostic interviews and self-report instruments. *Psychooncology*. **23**:121-130.
4. Hartung, T. J., Brähler, E., Faller, H., Härter, M., Hinz, A., Johansen, C., Mehnert, A. (2017). The risk of being depressed is significantly higher in cancer patients than in the general population: Prevalence and severity of depressive symptoms across major cancer types. *European Journal of Cancer, 72, 46–53*.doi:10.1016/j.ejca.2016.11.017
5. World Health Organization (WHO): Cancer. (2018) Retrieved from https://www.who.int/cancer/en/
6. Penedo, F. J., & Cella, D. (2016). Responding to the quality imperative to embed mental health care into ambulatory oncology. *Cancer, 123(3), 382–386*.doi:10.1002/cncr.30402
7. Lowery AE, Holland JC. (2011) Screening cancer patients for distress: Guidelines for routine implementation. Community Oncol. **8**: 502–5.
8. Smith, H. R. (2015). Depression in cancer patients: Pathogenesis, implications and treatment. *Oncology letters*, *9*(4), 1509-1514.
9. Howell D., Alici Y. (2010) Management of distress in cancer patients. *J Support Oncol*. 8: 4-12.

For peer review only

10. Mausbach, B. T., & Irwin, S. A. (2017). Depression and healthcare service utilization in patients with cancer. *Psycho-oncology*, *26*(8), 1133-1139.

11. Dauchy, S., Dolbeault, S., & Reich, M. (2013). Depression in cancer patients. *EJC Supplements*, *11*(2), 205.

12. Walker, J., Hansen, C. H., Martin, P., et al. (2014). Prevalence, associations, and adequacy of treatment of major depression in patients with cancer: a cross-sectional analysis of routinely collected clinical data. *The Lancet Psychiatry*, *1*(5), 343-350.

13. Vaibhav Kumar Ingle, Ijya Pandey, Akash Ranjan Singh, et al. (2017) Screening of Patients with Chronic Medical Disorders in the Outpatient Department for Depression Using Handheld Computers as Interface and Patient Health Questionnaire-9 as a Tool. Int J Appl Basic Med Res. 7(2): 129–133. doi: 10.4103/2229-516X.205809

14. "PA-18-493: Using Information Technology to Support Systematic Screening and Treatment of Depression in Cancer (R01 Clinical Trial Optional)." *National Institutes of Health*, U.S. Department of Health and Human Services, grants.nih.gov/grants/guide/pa-files/pa-18-493.html.

15. M.J. Hjermstad, H.C. Lie, A. Caraceni, *et al. (2012)* Computer-based symptom assessment is feasible in patients with advanced cancer: results from an international multicenter study, the EPCRC-CSA. J Pain Symptom Manage. 44; 639-654

16. Kingston D, McDonald S, Biringer A, et al. (2014) Comparing the feasibility, acceptability, clinical-, and cost-effectiveness of mental health e-screening to paper-based screening on the detection of depression, anxiety, and psychosocial risk in pregnant women: a study protocol of a randomized, parallel-group, superiority trial. Trials. 15:3. doi: 10.1186/1745-6215-15-3.

For peer review only

17. Aktas A, Hullihen B, Shrotriya S, Thomas S, Walsh D, et al. (2013) Connected Health: Cancer Symptom and Quality-of-Life Assessment Using a Tablet Computer: A Pilot Study. Am J Hosp Palliat Care. doi:10.1177/1049909113510963.

18. Gell NM, Grover KW, Humble M , *et al*. (2017) Efficacy, feasibility, and acceptability of a novel technology-based intervention to support physical activity in cancer survivors. *Support Care Cancer*2017;25:1291–300.doi:10.1007/s00520-016-3523-5

19. Quintiliani LM, Mann DM, Puputti M, Quinn E, Bowen DJ. Pilot and feasibility test of a mobile health-supported behavioral counseling intervention for weight management among breast cancer survivors. JMIR Cancer. 2016;2(1):e4.

20. Stinson, J. N., Jibb, L. A., Nguyen, C., Nathan, P. C., Maloney, A. M., Dupuis, L. L., et al. (2015). Construct validity and reliability of a real-time multidimensional smartphone app to assess pain in children and adolescents with cancer. *Pain, 156*(12), 2607–2615. doi: 10.1097/j.pain.0000000000000385.

21. Christ, G. H., Messner, C., & Behar, L. C. (2015). *Handbook of oncology social work: Psychosocial care of the patient with cancer*. New York: Oxford University Press.

22. Using PICO or PICo. (2019) Retrieved from https://libguides.murdoch.edu.au/systematic/PICO.

23. Dent, O. (2003). Methodological index for non-randomized studies. *ANZ Journal of Surgery,73*(9), 675-676. doi:10.1046/j.1445-2197.2003.02762.x

24. Higgins, J. P. T. (2020). Cochrane handbook for systematic reviews of interventions. Hoboken, NJ: Wiley-Blackwell.

25. Holland JC, Andersen B, Breitbart WS, et al. Distress Management. *J Natl Compr Cancer Netw*. 2013;11:190-2009. doi:10.6004/jnccn.2013.0027.

For peer review only

26. Mitchell AJ, Vahabzadeh A, Magruder K. Screening for distress and depression in cancer settings: 10 lessons from 40 years of primary-care research. *Psychooncology*. 2011;20(6):572-584. doi:10.1002/pon.1943.

For peer review only

Pubmed Search Strategy

((((Distress[Title/Abstract] OR depression[Title/Abstract])) AND screening) AND technology) AND cancer

((("Distress"[Title/Abstract] OR "depression"[Title/Abstract]) AND ((((((((((("diagnosis"[MeSH Subheading] OR "diagnosis"[All Fields]) OR "screening"[All Fields]) OR "mass screening"[MeSH Terms]) OR ("mass"[All Fields] AND "screening"[All Fields])) OR "mass screening"[All Fields]) OR "early detection of cancer"[MeSH Terms]) OR (("early"[All Fields] AND "detection"[All Fields]) AND "cancer"[All Fields])) OR "early detection of cancer"[All Fields]) OR "screen"[All Fields]) OR "screenings"[All Fields]) OR "screened"[All Fields]) OR "screens"[All Fields])) AND ((("technology"[MeSH Terms] OR "technology"[All Fields]) OR "technologies"[All Fields]) OR "technology s"[All Fields])) AND (((((((("cancer s"[All Fields] OR "cancerated"[All Fields]) OR "canceration"[All Fields]) OR "cancerization"[All Fields]) OR "cancerized"[All Fields]) OR "cancerous"[All Fields]) OR "neoplasms"[MeSH Terms]) OR "neoplasms"[All Fields]) OR "cancer"[All Fields]) OR "cancers"[All Fields])

Translations

screening: "diagnosis"[Subheading] OR "diagnosis"[All Fields] OR "screening"[All Fields] OR "mass screening"[MeSH Terms] OR ("mass"[All Fields] AND "screening"[All Fields]) OR "mass screening"[All Fields] OR "early detection of cancer"[MeSH Terms] OR ("early"[All Fields] AND "detection"[All Fields] AND "cancer"[All Fields]) OR "early detection of cancer"[All Fields] OR "screen"[All Fields] OR "screenings"[All Fields] OR "screened"[All Fields] OR "screens"[All Fields]

technology: "technology"[MeSH Terms] OR "technology"[All Fields] OR "technologies"[All Fields] OR "technology's"[All Fields]

cancer: "cancer's"[All Fields] OR "cancerated"[All Fields] OR "canceration"[All Fields] OR "cancerization"[All Fields] OR "cancerized"[All Fields] OR "cancerous"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields] OR "cancers"[All Fields]

For peer review only

PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Page number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	NA
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	3
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☐	NA
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐	☐	NA
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	13
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☐	NA
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	3-5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	5

For peer review only

BioMed Central
The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Page number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	5-6
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	7 (Supplemental File 1)
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	7
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	7
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	7
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	9-10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	8-9
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	9-10
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	9-10
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☐	NA
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☐	NA

For peer review only

BioMed Central The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Page number(s)
			Yes	No	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☐	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	10

For peer review only

BioMed Central
The Open Access Publisher